37. How do you use Data and AI affecting your future career or everyday life? Be specific with examples

Data and AI will have a huge impact on my future life and carcer, especially in logistics and Supply chain

In my role Data can help improve:
- Delivery planning
- Stock management
- Decision Making

AI can also help by:
- Predicting Demand
- Reducing Errors and Improving Efficiency

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SECTION E: Personal Reflection

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The concept I found most interesting is Artificial Intelligence (AI) especially how it can learn from Data and make decisions
I was surprised by how AI is already used in everyday life, such as voice assistance and recommendations on applications (APPS), Before this assignment, I didnt realise how much AI is part of things we use daily.

I also found it interesting that AI can improve over time by learning from data. This is a demonstration how powerful Data is in today's world

36. Which concept did you find most difficult to understand? What resources did you use to help you understand better?

- The concept I found ~~interesting~~ most difficult to understand was Data Lakes and how they are different from Data Warehouses

At first both seemed the same because both store Data however after doing more research, I understood a Data warehouse stores clean and structured Data, while Data Lakes stores raw and unstructured Data
- To understand better I used Google Search
- beginner explanations from IBM and Microsoft websites
- Youtube videos explaining Data Storage in simple terms

34. What is Business Intelligence (BI)? Name two popular Business Intelligence Tools (eg Power BI and Tableu) and explain how they help businesses make Data driven decisions.

Business Intelligence means using Data to help a business make better decisions

BI tools turn data into
- Reports
- Dashboards and Visual Arts

Example of BI tools:
- Power BI → used to create Dashboards and Reports

- Tableu
Used to visualise and Analyze Data

- Why is Business Intelligence (BI) important
  - ▷ Helps companies understand performance
  - ▷ Supports better decision making

32. What is Bias in AI? Explain what is means for an AI model to be biased. Provide a real world example of AI bias and describe the potential consequences.

- Bias happens ~~when~~ in AI when a system gives unfair results
  - This usually happens ~~when~~ because of poor or unbalanced data

Example:
A hiring system trained only on one group of people may:
- Prefer certain candidates unfairly

Consequences
- Discrimination
- Wrong decisions and loss of trust

33. What are the key differences between a Data Analyst, Data Scientist, Data Engineer and an AI Engineer? Create a simple table comparing their roles, skills and tools

Roles ~~What they~~ ~~What~~ ~~What~~ They Do		Skills	Tools
Data Analyst	Looks at Data and Creates Reports	Excel, SQL	Excel, Power BI
Data Scientist	Builds Models and Predictions	Python and Statistics	Python, Machine Learning Tools
Data Engineer	Builds Data Systems and Pipelines	Programming & Databases	SQL, Cloud Tools
AI Engineer	Builds AI systems	Machine Learning & Coding	Python, AI frameworks

Examples of Unethical use

- Using personal data without permission
  - This is wrong because people did not permit/agree to share their data

- Sharing Selling customer data secretly
  - This breaks trust and can harm customers

- Using AI to spread fake information
  - This can mislead people and cause harm

31. What is Data Governance? Define it and explain the key components of a good governance framework

- Data Governance is how a company or organisation manages its data and how it controls it.
  - It ensures data is: Accurate, secure and used properly

Key Components

- Data Quality
  - Data must be correct and reliable

- Data security
  - Data must be protected from hackers

- Data Policies
  - Rules on how Data is used

- Data Ownership
  - Knowing who is responsible for the Data

Common Python Libraries:
- Pandas: – Used for working with Data tables
- Numpy: – Used for Calculations
- Matplotlib – Used for Graphs and Charts
- Scikit-learn – Used for machine learning

SECTION D: Data Ethics, Governance and Careers

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to

- Data privacy means protecting peoples personal information so that it is not misused and this includes: Names, ID numbers and bank details
- Data privacy is important because: It protects people from fraud and identity theft
- It builds trust between companies and customers

- What is GDPR? – Its a law that protects personal data, it applies mainly to countries in the European Union [EU], but also affects companies worldwide that handle (EU) data

30. What is Data Ethics? Provide three (3) examples of unethical use of Data or AI, and explain what makes each example problematic?

Data Ethics means using Data in a fair, honest and responsible way

- A commit
This is saving your changes.

- A branch
This is a separate version of your work.

27. What is a Jupyter Notebook? What programming language is it commonly used with? Why is it popular among Data Analysts and Scientists

- A Jupyter Notebook is a tool use to write and run codes especially for Data Analysis

- It allows you to: Write code, See results immediately and Add notes and explanations
- It is mostly used with Python and popular because its easy to learn and use

28. What is Python? Why is it considered the most popular programming language for Data analytics? Name (4) four Python libraries commonly used in Data work and Briefly describe each.

- Python is a programming language used for Data analytics and AI
- It is popular because: Its easy to read and write
  - Powerful for Data work

25. What is Cloud Computing? Name the three (3) main cloud service models (Iaas, Paas, Saas) and Briefly explain each. Give one example of a product for each model

· Cloud computing means using internet to store data and run applications (app) instead of using your own computer

Main types

• Iaas (Infrastructure as a Service)
You rent servers
Example: Amazon Web services

• Paas (Platform as a Service)
You can use a platform to build apps
Example: Google Cloud

• Saas (Software as a Service)
You use software online
Example: Microsoft 365

26. What is a Version Control System? What is Git and why is it widely used by Data analysts and developers? Briefly explain what a 'Commit' and a 'Branch' are

· A version control system tracks changes made to files over time.

It helps developers
• Keeps history of changes
• Work together

· Git is the most popular version control system

23. What is Prompt Engineering? Why it is an important skill when working with AI tools like Chat GPT or Claude? Provide 2 examples of a poorly written prompt and an improved version of each.

- Prompt engineering is a skill of writing clear instructions for AI tools

Its important because better prompts = better answers
Example:
Poor prompt = Explain Data
Better prompt = Explain Data Analytics in simple terms with examples
Poor prompt: Help me
Better prompt: Help me understand databases like I am a beginner with examples

24. What is a MAP function? Explain what it does in programming and Data processing: Provide a simple example in either Python or JavaScript

A MAP function is used to apply the same action to every item in a list
It changes each item one by one
Example in Python

```
numbers = [1, 2, 3]
result = list(map(lambda x: x*2, numbers))
```

Output: [2, 4, 6]

10 Section C: APIs, Tools and Developer Concepts

19. What is an API ~~Key~~ (Application Programming Interface
Explain what it is using a simple real world analogy
Why are APIs important in modern software and Data Systems

An API is a way for different Apps ~~for~~ systems to talk to each other

Example: An API is like a waiter in a restaurant

- I (User) places an order
- Waiter (API) takes it to the Kitchen (System)
- Food (Data) comes back to me

APIs are important because they allow systems to share data easily

20. What is an API Key? What is its Purpose? How should it be stored and Protected and What could happen if its exposed publicly?

An API Key is a secret code used to access an API

It is used to: Identify who is using the API
Control access and security

API Keys must be: Kept private and Not shared publicly

If exposed someone can Use your API without permission
Cause security problems or extra costs.

18. What is Computer Vision? How do machines 'see and interpret images'? Provide two real world use cases of computer vision

Computer vision is a type of AI that allows machines to see and understand images or videos, machines do this by analysing pixels and patterns

Examples: Face recognition on smartphones
• Self driving cars detecting roads and objects

16 What is Generative AI? How does it work at high level? Name three tools that use Generative AI and describe what each one does?

Generative AI is AI that can create new content like text, images or music, it works by learning patterns from existing data and generating something new

Examples of Tools:
- Chat GPT — creates text answers
- DALL-E - creates images
- GitHub Copilot - helps write codes

17 What is a Training Data Set? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect.

A training dataset is the Data used to teach an AI Model, the quality of this data is very important
If the Data is incorrect:
- AI gives wrong answers
- biased — AI becomes unfair

Example: If an AI is trained for one group of people, it may not work for others

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real world applications of NLP that you use in everyday life.

NLP is a type of AI that helps computers understand and use human language

Examples in daily life:
- Voice assistants like Google Assistant
- Chat Apps like WhatsApp (Auto suggestions)
- Translation tools like Google Translate

15. What is Large Language Model (LLM)? How is it different from standard Machine Learning models? Name well known LLMs and Explain what they are used for?

A Large Language Model is an AI system trained on a lot of text to understand and Generate language
It can answer questions, write text, and explain ideas

Difference from normal ML:

- LLMs handle language better
- They are trained on much larger data sets

Examples: Chat GPT and Claude

5 SECTION B (Q11-18)

11. What is Artificial Intelligence (AI)? Define and Explain
• Artificial intelligence (AI) is when computers are made to think and Act like humans
AI can learn, solve problems, and make decisions
There are two types of AI

Narrow AI (Weak AI)
This type does one specific task only
Example Siri or Netflix recommendations

General AI (Strong AI)
This type can think and do anything like a human
It does not fully exist yet

12 What is Machine Learning? How does it differ from traditional programming? Describe the 3 main types of machine learning: Supervised, Unsupervised, and Re inforcement learning.

Machine Learning is a type of AI where computers learn from Data instead of being programmed step by step

Difference from traditional programming

• Traditional programming → You give it rules
• Machine Learning → Computer Learns patterns from Data

2

4 What is a Data Mart? How does it relate to a Data ~~Mart~~ warehouse and who typically uses it?
A data Mart is a small part of a Data Warehouse focused on one department, Data wave house is like the whole company data but Data Mart is a department's data. Who use's it, eg Logistics, Finance or Sales Team

5. Explain the difference between Structured, Semi-structured and Unstructured Data. Provide one example of each.
Structured, Unstructed and Semi-Structured Data
> Structured – Tables eg Excel sheet, SAP data
> Semi structured – some structure but not perfect eg Emails
> Unstructured – No structure at all eg Videos, images and Voice Notes

6. What is Data Modelling? Why is it important when designing a system that works with data? Describe at least two types of Data Models.
Data Modelling is planning how data will be organised before building a system, like drawing a plan before building a house. it keeps data clean and structured, makes system faster and easier to use. Conceptual Model: big picture Model ② Logical Model: More detail

7. What is Data Mining? How is it used in Business? Provide two realk world examples of Data Mining in action
Data mining means finding useful patterns in Data
it helps companies understand customers and helps predict future trends
Examples: Retail Stores: find what products people buy together
Banks: Detect fraud transactions

8. What is Database Management System [DBMS]? Name and briefly describe four (4) popular DBMS platforms (e.g MySQL, PostGreSQL, Oracle, MongoDM) and explain a suitable use case for each

A Database Management System (DBMS) is a software used to create and manage database

1. MySQL: Easy and Widely used; Use small to medium apps
2. Postgre SQL
   - More advanced features
   - Use complex systems
3. Oracle Database
   - Used by Big Companies
   - Use: banking systems
4. Mongo DB
   - Stores flexible data (Not tables)
   - Use: apps with changing Data

9. What is ETL (Extract, Transform and Load)? Explain each step and describe why this process is important in data analytics

Extract: Take data from different sources like Excel, SAP or even emails

Transform: Clean and Fix data: Remove errors, format properly

Load: Store it on a database or data warehouse

Why is it important: Makes data clean and usable and helps with accurate reporting

Example: Collecting diesel delivery data → Clean it → Load into reporting system. All my examples are based on my JOB as a Logistic Coordinator Diesel & Petrol deliveries.

Page 1

Foundations of Data Analytics & Data Science

1. What is a Database? Define in you own words and Explain why databases are essential in the morden world. Provide two reall ~~ex~~ world examples of where databases are used.

Database is a place where a computer stores information in an organised, structured way, like a table with rows and columns
Why are they important. They keep information neat and easy to find
You can change and update data quickly
Businesses use them to make decisions

Eg.
Bank system - Stores / Keep your account balance and transactions
Shop system - Stores like Pick n Pay stores Product, price and Stock

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a Data Warehouse?

Data wave house is a big storage system that collects / gather data from many places and stores it for analysis
DW - uses Online Analysis Processing (for analysis and Reporting)
DB - uses Online Transactional Processing (Daily use) (OLTP)
DB - used daily transactions, small and fast
DW - Large and Historical, used for analysis

3. What is a Data Lake? Explain the concept and describe ~~house~~ how it differs from a Data warehouse. When would a business prefer a data Lake over a data Warehouse?

A data lake stores all types of Data, even messy unorganised data eg. Data wave house is like a clean fridge
Data lake everything is dumped in one big container

Differences

Data wave house	Data Lake
Clean, Structured organised Data	Raw messy Data
Ready for reporting	Needs to be processed first
eg. Storing Videos, emails etc.	

10. What are common data storage formats used in analytics (eg CSV, JSON, PARQUET, Avro)? Briefly describe each and state when you would use one over another

1. CSV (Comma Separated Values)
- Simple text file
- Looks like excel

Use: small data, easy sharing

2. JSON (Java Script Object Notation)
- Stores data in key value format

Example {
"name": "Diesel",
"volume": 40

Use: APIs and Web apps

3. Parquet
- Fast and Compressed format

Use: Big data
- Analytics systems

4 Avro
- Stores data with structure + schema

Use: Data streaming systems